

Role of the Local Electric-Field in High-Field Conditioning of DC Electrodes: Numerical and Experimental Insights

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Conditioning, the progressive increase of voltage-holding through the controlled application of fields, is an important and widely used process for bringing high-field and high-voltage devices up to their full operating parameters. Here, a study is presented on how conditioning can vary within a device, specifically, when there is a spatial variation in the surface electric field. What has been observed in high-field pulsed direct-current electrodes and radio-frequency structures is that locations exposed to higher fields exhibit a greater tendency to breakdown, but this increase is counteracted by an increased conditioning rate. This interplay explains the observed breakdown locations and provides important insights into the mechanisms underlying both conditioning and breakdown. This study combines Monte Carlo simulations with experimental results from pairs of high-field electrodes with a radially varying surface electric field. Results are presented from a high-field pulsed DC system, in which the position of each breakdown during conditioning was recorded by triangulation using a pair of cameras, and the results are compared with Monte Carlo simulations.

I. INTRODUCTION

In many high-field and high-voltage devices, it is common practice to gradually increase the operating voltage while monitoring for arcs, rather than immediately commencing operation at the design voltage. Doing so typically increases the likelihood of establishing high electric fields without arcing. This procedure, known as conditioning, is an essential step in the commissioning process for vacuum interrupters and normal-conducting particle accelerator cavities and waveguide components [1–3].

The investigation of field-driven conditioning has both practical and theoretical benefits. Conditioning procedures are time-intensive processes and typically carry an inherent risk of component damage. An improved understanding of the process can aid in identifying material, manufacturing, and treatment choices that mitigate the need for conditioning, reduce associated risks, and facilitate optimization of existing procedures. A prominent example is the high-gradient radio-frequency (RF) cavities developed for applications ranging from cancer treatment to future collider concepts, i.e. flash therapy and CLIC [4–6]. Cavities of this kind operate at surface fields up to ~ 220 MV/m, and generally require substantial conditioning periods [5–8].

To shed light on the mechanisms that underlie conditioning, a Monte Carlo simulation was previously developed at CERN [9]. The report presented here describes the design of two electrodes based on simulations performed with this tool, followed by a dedicated experiment

conducted in the CERN pulsed DC system [10, 11]. To investigate the field dependence of conditioning, a tapered electrode design was employed to establish an inhomogeneous surface electric field. The electrodes were conditioned to a peak surface electric field of approximately 80 MV/m, and the breakdown positions were recorded on a pulse-to-pulse basis. The spatial variation in the surface electric field, combined with precise temporal and spatial breakdown data, serves two main purposes. Firstly, the results shed light on the mechanisms underlying conditioning and breakdown by clearly illustrating the interplay between the applied field that produces a conditioning effect and the field that drives breakdown, and by providing information on the relative strength of each phenomenon. Secondly, the results serve as an important dataset against which models of conditioning and breakdown, such as the one presented herein, may be benchmarked.

II. FEATURES OF HIGH-FIELD CONDITIONING

To date, numerous experiments have been conducted to explore the mechanisms associated with high-field vacuum arcs, field holding, and conditioning. Parameters that have been studied include the device geometry, material, preparation techniques, operating frequency (including DC), the gaps between electrode surfaces, and the high-field pulse length and repetition rate [12–17]. Recently, DC and RF tests have also been performed at cryogenic temperatures, and this approach now forms the basis of a proposed future collider concept [18–22]. Unlike normal-conducting structures, where RF conditioning is commonly employed to reduce breakdown rates within

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the accelerating structure itself, superconducting cavities rely primarily on meticulous surface preparation and contamination control to suppress breakdown-inducing mechanisms. RF conditioning is therefore generally applied only to the input couplers as part of its acceptance test [23, 24]. In the context of existing facilities, attempts have been made to discern the effects of radiation exposure on high-field performance [25, 26].

A comprehensive set of models that describe the different experimental signatures, dependencies, and stages of breakdown has been developed. The breakdown process first involves material dislocation dynamics driven by the stress from the applied surface electric field, followed by the formation of surface features, their evolution driven by surface dynamics and finally heating, electron emission and field driven neutral atom emission and finally plasma formation. A review of the current status of the understanding of the breakdown process is given in [27]. An improved understanding of conditioning will contribute to further progress in understanding breakdown.

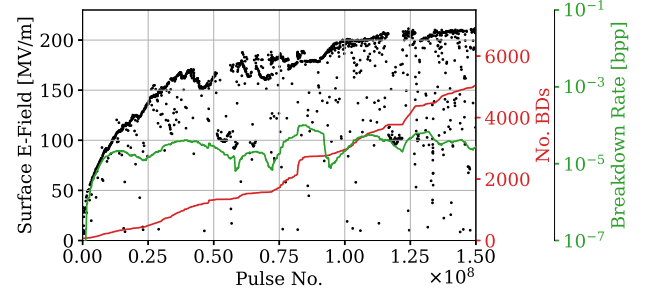
A. Conditioning Procedures

For the purposes of this work, conditioning is defined as the increased propensity to establish high fields without arcing or increases in internal pressure that would prohibit operation in normal-conducting vacuum devices. In the early stages of operation, some high-field devices, particularly RF cavities, are commonly limited by multipacting and elevated pressure levels rather than by arcing alone [28–31]. In one case, an algorithm was developed specifically to condition cavities using the pressure level as the process variable [32]. However, after this period, the rate at which the operating field can be increased is often limited primarily by the occurrence of vacuum arcs. Historically, conditioning procedures were often carried out on anecdotal or empirical evidence rather than a precise scientific understanding of the process. However, automated algorithmic approaches to conditioning are now commonplace, and offer a more consistent and systematic approach to component testing and commissioning [2, 32–35].

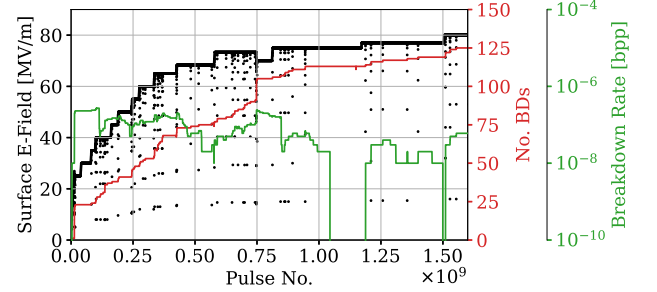
Generally, conditioning procedures, both manual and algorithmic, are based on the breakdown rate, i.e., the number of vacuum arcs that have occurred within a chosen temporal window, or equivalently, the number of pulses. The pulse width and field level are then adjusted to maintain the target breakdown rate until the final operating parameters are reached. Practically speaking, this means progressively increasing the operating field while monitoring for arcs to maintain a permissible rate of arcing. However, additional action is also occasionally required to respond dynamically in the event of, e.g., an elevated arc rate or otherwise anomalous behavior.

B. Characteristics of Conditioning

During conditioning procedures, if a constant breakdown rate is maintained, the rate at which the field can be increased typically decreases, leading to asymptotic behavior. To illustrate this behavior, the conditioning of an X-band accelerating structure and a DC electrode are shown in Fig 1. The approach of progressive field ramping has been shown to reliably enable operation at high fields with a reduced probability of arcing, and a major theoretical objective of this work is to gain insight into the mechanism underlying this behavior.



(a) The conditioning curve of a 12 GHz copper RF cavity (T24 PSI N1 [17]).



(b) The conditioning curve of a copper electrode (**NAME?**) that was tested in the LES at CERN.

Figure 1: Conditioning curves of a 12 GHz copper RF cavity (T24 PSI N1 [17]) that was tested in CERN's high-gradient test facility (a) and a copper DC electrode (**NAME?**) that was tested in the LES with a gap size of 60 μm (b), illustrating a typical asymptotic increase in applied field. Note that the x-axes differ in scale by an order of magnitude.

Long-term tests performed on X-band ($\sim 12\text{ GHz}$) RF cavities have shown that the conditioning of similar devices proceeds most comparably in terms of cumulative high-field pulses rather than the cumulative number of breakdowns [36]. This result has led to the suggestion that the conditioning effect, i.e. the increased field-holding capability, is produced by the application of high fields and is not solely a consequence of the arcs. Therefore, DC electrodes are conditioning in a staircase fashion, where the applied electric field is increased more gently, see Fig. 1 (b). The conditioning approach is explained

in more detail in Section III. This result forms the basis for much of the work conducted in this manuscript and is discussed further in Section IV.

C. Characteristics of Vacuum Arcs

As they constitute one of the main limitations on stable high-field operation, significant effort has gone into improving the understanding of vacuum arcs, as well as predicting and mitigating their occurrence [37]. In various RF and DC high-field tests, it has been shown that if an arc occurs following a quiescent period, it is common for one or several more arcs to occur in quick succession, and that these groups of events are likely to occur in close proximity [11]. This observation has led to the grouping of arcs into primary or trigger events, which occur stochastically, and secondary events, which are thought to be consequences of the former [11, 17, 38].

To date, several quantities have been studied as predictors of device performance. Examples include the group velocity in RF devices [39, 40], the ratio of the power flow to the iris size in accelerator cavities [41], and the so-called modified Poynting vector, S_c [42]. Although RF cavity studies have shown that, for a given breakdown rate, similar peak magnetic fields are obtained for different designs [14], post-test optical examinations have shown that the arcs predominantly occur where the surface electric field is highest [43]. An example is shown in Fig. 2, where the surface of an X-band RF cavity that operates in the TM_{110} mode is presented with the surface electric field distribution and the arc locations overlaid. In this mode, the regions where S_c , the surface electric field, and the surface magnetic field are well separated, and a post-test optical examination showed that the majority of breakdown sites were present where the electric field is highest [43].

More broadly, a comprehensive analysis of long-term

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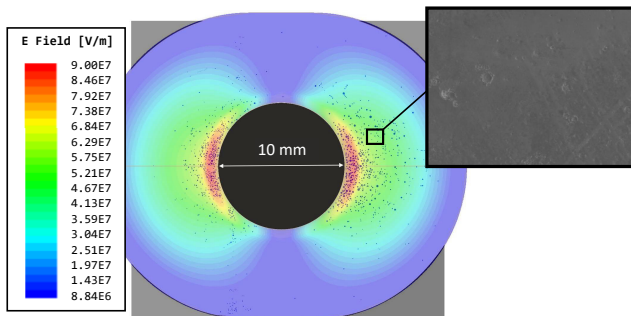


Figure 2: Electric field distribution on the second cell of the CLIC crab cavity with the breakdown locations superimposed (shown by the dark blue spots). Figure taken from a previous publication [9]. The test procedure and post-test analysis are reported in full elsewhere [43].

RF cavity tests found the breakdown rate to scale with the electric field E , the thirtieth power, or E^{30} [42]. More recent tests have shown similar power law relations, with reported exponents ranging from 28 to 35 [17, 44]. However, this scaling emerges globally, and weaker dependencies have been reported across regions of inhomogeneous field within a given device [43], as shown in Fig. 2. In pulsed RF devices, the breakdown rate is commonly quoted as scaling with the pulse length, τ , to the fifth power, τ^5 [36, 42]. However, recent attempts to measure this dependency have yielded lower exponents, and the interpretation of such measurements is occasionally complicated by the use of non-rectangular pulse shapes [17, 45, 46]. In a previous study, the potential influence of the order in which the measurements are conducted was also noted [17].

D. Motivation

Generally, arcs serve as the feedback mechanism during conditioning procedures, with the occurrence of an arc indicating that the field level is sufficiently high to produce a non-negligible conditioning effect. During such procedures, the field can gradually be increased while maintaining a constant rate of arcing. Conversely, if the applied field is held constant, measurements have shown that the arc rate gradually decreases. These observations suggest that the application of high electric fields increases the likelihood of arcing and that the arc rate and/or the conditioned state of the surface continuously evolve over the course of a given measurement. The objective of this work is to investigate the interplay between these two effects and to provide information on the relative strength of each mechanism.

In multicell traveling wave (TW) RF cavity tests, localization of arc sites to a specific cell is often accomplished in situ by employing a time-of-propagation-based method [47]. However, localization to a specific region on the surface is generally not possible without a post-test visual examination. Consequently, the exact spatial and temporal evolution of the arc distribution on the surface is lost. In some conditioning studies using DC electrodes, precise real-time breakdown localization is possible, though only samples that support homogeneous electric fields have been tested thus far [48]. In this study, this limitation is addressed by testing a pair of electrodes that establishes an inhomogeneous surface electric field while recording the arc locations and the order in which they occur. The electrodes were designed with the aid of a Monte Carlo model of conditioning [9]. The objective of the optimization process is to select a geometry that, after conditioning, would exhibit a gradual decrease in arc density across its surface.

Therefore, to the authors' knowledge, this dataset represents the first time that the positions of arcs and the order in which they were accrued have been recorded simultaneously in a pulsed device operating with an

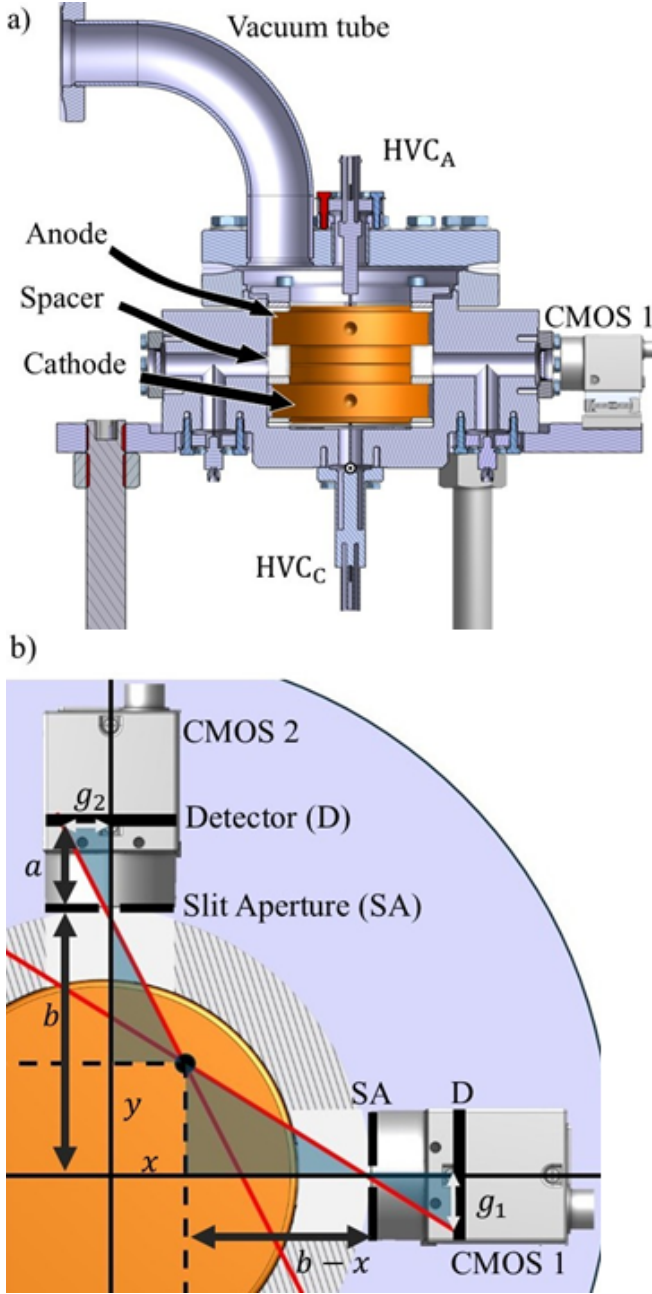


Figure 3: Cross-sectional views of the LES shown from the side (a) and from above (b). a) The electrodes are separated by an insulating spacer, creating a micrometer precision gap. Electrical contact to the anode and cathode is made via the HVC_A and HVC_C , respectively. Two CMOS cameras are attached to the four symmetrical viewports on the system. Only one is shown here, as the other is hidden behind the system. b) Breakdown localization system. When a breakdown occurs, it creates a flash of light originating from (x, y) . Two red lines can be drawn, going through the vacuum breakdown and the slit apertures (SA), and hitting the detectors (D). This creates two similar triangles, marked in blue, from which the original breakdown location can be calculated using the known system constants a and b and the measured quantities g_1 and g_2 . The equations are given elsewhere [13].

inhomogeneous surface electric field. The results also serve to validate the model used to design the electrodes, thereby confirming its suitability for more pragmatic applications, such as optimizing existing conditioning procedures.

Notably, at the High Energy Accelerator Research Organization (KEK), a camera was integrated into a high-power cavity test, allowing direct in-situ observation of breakdowns as they occurred [49]. However, this arrangement was operated in a continuous wave regime and focused on mechanisms associated with breakdown triggers, rather than on the evolution of the arc distribution and the conditioned state of the surface, which are the focus of this work.

III. EXPERIMENTAL SETUP, CONDITIONING PROCEDURE AND SPATIAL LIGHT TRIANGULATION

The experimental measurements were performed in a vacuum chamber fitted with a series of interchangeable electrodes. Due to the large size of the electrodes, this configuration is commonly referred to as the Large Electrode System (LES) [10, 50–53]. Figure 3 a) shows a cross-sectional view of the setup. Inside, the two electrodes are separated using a precision-machined insulating spacer. The spacer rests on the cathode, while the anode sits atop the spacer. Two pin-type high-voltage connector (HVC) feedthroughs, specialized for delivering and receiving power to and from the interior of a vacuum chamber. HVC_A and HVC_C connect to the anode and cathode, respectively. On the side, one of the two CMOS cameras is shown. The second is positioned perpendicular, at the back of the chamber.

The chamber is evacuated using a single turbomolecular pump to achieve vacuum levels down to 10^{-9} mbar. Pressure is monitored along the vacuum line using a cold-cathode vacuum gauge and recorded at a sampling rate of 3 Hz.

The electrodes typically comprise a flat machined ($R_a < 20$ nm) metallic cathode and anode to ensure a spatially uniform surface electric field. Depending on the chosen spacer, the typical electrode gaps are 20, 40, 60, or 100 μm . Here, the 60 μm spacer was chosen because it is a common gap size. The gap distance is verified by capacitance measurements, as the electrodes act as a parallel-plate capacitor. The exposed radii of the cathode and anode are 30 mm and 20 mm, respectively. These dimensions remove edge-related field enhancement, which would localize breakdown events at the electrode periphery due to the increased surface electric field [50]. The electrodes consist of oxygen-free (OFE) Cu electrodes, which were heat-treated in a hydrogen atmosphere up to 800 $^{\circ}\text{C}$ to extend the conditioning time [54].

To achieve a non-uniform electric field, the anode was machined to have a flat-topped conical shape, a frustum. The overall gap was determined by the spacer used. This

leads to the gap height $d(r)$ that is a function of the radius

$$d(r) = \begin{cases} 60 \text{ } \mu\text{m}, & r \leq 6.5 \text{ mm} \\ 60 \text{ } \mu\text{m} + (r - 6.5 \text{ mm}) \tan(\theta), & r > 6.5 \text{ mm} \end{cases} \quad (1)$$

where $\theta = \tan^{-1}((60 - h_2) \text{ } \mu\text{m}/23.5 \text{ mm})$. The gap height at the anode edge, h_2 , was determined by simulations and is covered in detail in Section IV. Consequently, the surface electric field can be found as

$$E_{\text{surface}}(r) = \frac{V_{\text{applied}}}{d(r)}, \quad (2)$$

ensuring a uniform surface electric field when $r \leq 6.5 \text{ mm}$ and a radially decaying when $r > 6.5 \text{ mm}$.

During testing, short high-voltage pulses are applied to HVC_A provided by a pulsed high-voltage generator (EPULSUS-FPM1-10) [55], while HVC_C is grounded. The generator employs a Marx circuit configuration, charging ten capacitors in parallel before discharging them in series, hence the name Marx generator. The LES Marx generator can deliver pulses up to 10 kV and 50 A, with pulse widths ranging from 500 ns to 100 ms. The generator is powered by an external supply.

When a vacuum breakdown occurs during a pulse, the chamber transitions from an insulating vacuum between the electrodes to a low-impedance state, causing a rapid surge in current demand of up to 50 A within 10 ns. The breakdown is detected by the sudden current demand from the Marx generator, which prompts the applied pulse to stop after 600 ns. During this, the pressure will spike to the order of 10^{-6} mbar before decaying exponentially back to stable values. Therefore, the system will wait 3 s before pulsing continues to let the vacuum recover. During recovery, the applied voltage is reduced to 20 % of its pre-breakdown value and then logarithmically ramped back up to the conditioning set-point [56]. If the breakdown rate (BDR), the number of breakdowns per pulse, exceeds a preset threshold (typically 10^{-5} bds/pulse), the system actively lowers the applied voltage until the BDR is met [56]. A collection of conditioning studies using this experimental setup can be found in the literature [25, 26, 54, 56, 57].

The vacuum breakdown also produces brief flashes of light that resemble a point source and persist for only a few microseconds [58]. For copper, the light emission in the LES occurs as a broadband signal across the visible spectrum, with its peak at 750 nm [56]. Two perpendicular cameras, each equipped with a 100 μm wide slit aperture, are gathering the optical data from the chamber, as seen in Fig. 3 b), operating continuously and synchronized with an exposure time of $\tau = 3 \text{ s}$.

The slit apertures restrict the transmitted light so that only the central portion of the optical beam reaches the detectors, producing a Gaussian intensity profile on each camera. A line drawn through the center of the Gaussian profile and the corresponding slit aperture defines the optical line of sight, which also passes through the breakdown location. By having two cameras, the breakdown

location (x, y) can be found as the point at which the two lines cross, as shown in Fig. 3 b). The two measurable quantities, g_1 and g_2 , along with the system constants, a and b , are used to find the origin location with equations given in [10].

IV. SIMULATION AND DESIGN OF A TAPERED ELECTRODE

Previously at CERN, a Monte Carlo-based model was developed to simulate the conditioning and operation of high-field devices on a pulse-to-pulse basis [9]. The model is deliberately constructed to be theory-agnostic and relies on two main assumptions:

1. The application of an electric field produces a local conditioning effect, i.e., exposure to high electric fields increases the surface's propensity for establishing electric fields without arcing.
2. The probability of arcing scales strongly with the local electric field, with higher fields corresponding to an increased probability of arcing.

In addition to more pragmatic motivations, such as optimizing conditioning procedures, one of the key objectives was to investigate the interplay between these two counteracting effects and the discrepancies in empirical scaling laws that have been reported. Although the model, and several preliminary studies conducted with it, have been discussed elsewhere [9], it has since undergone several modifications and has not been applied to DC devices. As it was employed in the design of the electrodes reported in this paper, and therefore forms the basis for much of the work reported herein, an overview is provided. Several peculiarities specific to the simulation of copper electrodes operating in a pulsed DC regime are also discussed.

A. Overview of the Model

The surface of the high-field device to be simulated is first meshed, yielding an array of n elements. Taking i as the element number, each element has the following attributes:

- a_i - The ratio of the mesh element area to the total simulated area.
- k_i - The ratio of the local electric field to the peak electric field for a given operating voltage or power level.

With this approach, the high-field conditioning of any arbitrary geometry and surface electric field distribution is permitted. Additionally, the use of an inhomogeneous mesh permits the use of larger elements in regions where

Morten finds it strange to not introduce the final design here, I don't

NOTE TO LEE, MAYBE CELL IS A MORE APPROPRIATE TERM AS THESE AREN'T TET MESH?

little activity, e.g., infrequent arcs or little conditioning, is expected to reduce the computational cost of the simulation.

Numerous experiments worldwide have shown that the voltage or power level required to establish a given breakdown increases asymptotically during conditioning procedures. Based on this observation, the model assumes that for a given pulse length and reference breakdown rate, P_{Ref} , there is a maximum, material-dependent surface electric field, E_L , in MV/m that a device can be conditioned to operate at. More succinctly, this assumption forbids the unphysical procedure of conditioning to arbitrarily high field levels or low breakdown rates.

To capture the conditioning effect, each surface element is also a conditioning state, $E_{S,i}$, which refers to the field level in MV/m at which the mesh element is conditioned to operate. Each element is also assigned a state enhancement factor, ψ_i , which is initially selected randomly from an appropriately tailored Gaussian distribution, and reselected each time the element accrues an arc. Depending on the applied peak electric field, E_O , which also has units of MV/m, the conditioning state of the surface elements is increased on a pulse-to-pulse basis as

$$\Delta E_{S,i} = \gamma_i \cdot \frac{E_O \cdot k_i}{E_{S,i}} \cdot \left[1 - \frac{E_{S,i}}{E_{L,i}} \right], \quad (3)$$

where γ_i is a constant with units of V/m that allows the model to be fit to experimental data, and varies depending on the material, operating frequency, and pulse length. Generally, most breakdowns that occur during conditioning procedures occur in the preliminary stages, as the applied field is increased to the target level [17, 36]. As the pulse length is typically fixed during this preliminary stage, the use of a fixed value for γ_i is sufficient for the studies reported in this work.

Although some variation is present, in well-conditioned copper RF devices the breakdown rate has typically been observed to increase with the applied electric field to the thirtieth power, i.e., $BDR \propto E_O^{30}$, [17, 42, 44], and investigating whether this scaling accurately reproduces the behavior of other devices, particularly those constructed from different materials or with different operating frequencies, is another objective of the model. Based on this empirically derived scaling, the probability of a particular grid element arcing on a given pulse, $P_{BD,i}$, is expressed as

$$P_{BD,i} = a_i \cdot \left[\frac{E_O \cdot k_i}{E_{S,i} \cdot \psi_i} \right]^{30} \cdot P_G, \quad (4)$$

where P_G is a probability that is applied to the grid elements individually to obtain a global breakdown rate that is comparable to P_{Ref} in the later stages of conditioning. To derive a suitable value for P_G , the probability of a pulse not resulting in an arc is more convenient to

define. Assuming the system is fully conditioned everywhere, and all ψ_i values are equal to one,

$$1 - P_{Ref} = \prod_{i=1}^n (1 - P_G \cdot a_i \cdot k_i^{30}). \quad (5)$$

Although a solution can be obtained using iterative numerical methods, a reasonable approximation can also be obtained. Taking the natural logarithm of both sides of Eq. 5 yields

$$\ln(1 - P_{Ref}) = \sum_{i=1}^n \ln(1 - P_G \cdot a_i \cdot k_i^{30}). \quad (6)$$

However, $P_{Ref} \ll 1$, with typical values being less than 1×10^{-4} . Therefore, by taking the first term of the Taylor series, a reasonable first-order approximation may be obtained as

$$\ln(1 - P_{Ref}) \approx -P_{Ref}. \quad (7)$$

As $P_G \cdot a_i \cdot k_i^{30} < P_{Ref}$ for any scenario with more than one grid element, the same holds true for the right side of Eq. 6. Rearranging the terms then yields

$$P_G \approx \frac{P_{Ref}}{G}, \quad (8)$$

where G is a geometry factor, which corresponds to the surface integral of the surface electric field to the thirtieth power, defined

$$G = \sum_{i=1}^n a_i \cdot k_i^{30}. \quad (9)$$

Consequently, Eq. 4 dictates that if all ψ_i values are equal to one, every surface element has been conditioned to its limit, $E_{L,i}$, and the device is operated at this level, the breakdown rate will be approximately equal to P_{Ref} . This ensures that the performance in simulation is consistent with the experimental data from which $E_{L,i}$ and P_{Ref} were selected, regardless of the mesh resolution. In low-field regions, the assumption that the surface has been fully conditioned is not necessarily correct. However, the probability of breakdown scales extremely strongly with the applied field, as per Eq. 4, and so the error that this effect introduces is small.

In several previous studies, an alternative approximation was used [9], which instead relied on the assumption that

$$1 - P_{Ref} \approx (1 - P_G)^G. \quad (10)$$

Rearranging this expression, P_G can then be defined as

$$P_G = 1 - (1 - P_{Ref})^{\left(\frac{1}{G}\right)} \quad (11)$$

This approximation emerges from taking the natural logarithm of the left-hand side in Eq. 10, exponentiating the result, and, as $P_g \ll 1$, once again taking the first term of the Taylor expansion to show

$$\begin{aligned} (1 - P_G)^G &= \exp(\ln(1 - P_G) \cdot G) \\ &\approx \exp(-P_G \cdot G) \\ \therefore 1 - P_{Ref} &\approx \exp(-P_G \cdot G), \end{aligned} \quad (12)$$

and therefore, taking the natural log once again and employing the approximation given in Eq. 7, the same approximation as Eq. 7 emerges as

$$-P_{Ref} \approx -P_G \cdot G. \quad (13)$$

The definition in Eq. 11 holds if the variation in $a_i \cdot k_i^{30}$ products is small, and therefore is suitable for devices where the mesh and electric field distribution are uniform, or in cases where regions of lower electric field are compensated by much larger mesh cells. However, the form given in Eq. 8 yields an improved result when larger cell-to-cell variations are present, and is therefore more broadly applicable. For this reason, the Eq. 8 definition was employed for the studies presented in this work.

Regarding the selection of P_{Ref} and E_L , to accurately reproduce a device's behavior, the chosen values should ideally be taken from late-stage measurements conducted with the pulse length used in the early stages of conditioning. Measurements at multiple pulse lengths are often conducted during more comprehensive high-field tests [17, 59], and so this aspect of the model is generally not problematic. Additionally, as γ_i is presumed to be constant for a given set of device parameters, only a single appropriate test is required to derive an appropriate value.

Additionally, the P_{Ref} and E_L values are somewhat interchangeable, provided the γ value has been appropriately selected so as to provide the correct dynamic behavior. Hence, to tune the model, the device's approximate final performance must be known a posteriori. Once the material properties are known and verified, however, the model may be applied to other geometries and devices by simply rescaling P_{Ref} to account for changes in surface area and electric-field distribution in the new design. Taking a trivial example, if two identical reference devices were to be simulated in parallel, the surface area, and therefore the reference breakdown rate, should also double for a given field level. More generally, the chosen reference breakdown rate is adjusted to provide a scaled reference breakdown rate, P_{Ref}^S , as

$$P_{Ref}^S = \frac{A^S \cdot G^S}{A \cdot G} \cdot P_{Ref}, \quad (14)$$

where A^S and A are the surface area of the device being simulated, and the device from which P_{Ref} was taken, respectively. Similarly, G^S and G are the geometric factors, as per Eq. 9, of the device being simulated and the reference device, respectively. The probability applied to the grid elements of the new device, P_G^S , is then defined like in Eq. 8, as

$$P_G^S \approx \frac{P_{Ref}^S}{G^S}. \quad (15)$$

and hence, for the new device Eq. 4 becomes

$$P_{BD,i} = a_i \cdot \left[\frac{E_O \cdot k_i}{E_{S,i} \cdot \psi_i} \right]^{30} \cdot P_G^S. \quad (16)$$

B. Electrode Design

To provide a reasonable foundation for the design of an appropriate electrode, the model described in Section IV A was first tuned by eye to approximately reproduce a standard, flat electrode (**Mention gap size**) based on the results of previous high-field copper electrode tests conducted at CERN. A simulation of the conditioning of a copper electrode using the field ramp shown in Fig. 1 is shown in Fig. 4, and chosen model settings are summarised in Table I. The total number of breakdowns accrued in the simulation is lower than in the experimental case, as the anomalous cluster of breakdowns that occurred around 750 million pulses in Fig. 1 and necessitated a reduction in operating field during the experiment was not reproduced by the model.

A cross-section of the frustum electrode, including the relevant dimensions, is shown in Fig. 5. The LES system and existing spacers are designed to accommodate anodes with a radius r_0 of 20 mm and, as described in Section III, a gap size h_1 of 60 μm is a standard and commonly used configuration for electrode tests.

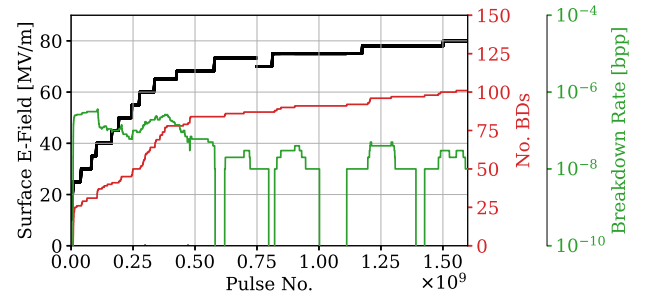


Figure 4: Simulated conditioning of a flat DC copper electrode. The model was tuned using existing LES data to produce similar behavior. The parameters employed are summarised in Table I. A total of 40 mesh cells were used.

Table I: The simulation settings used to approximately reproduce a copper cathode that was tested to a field level of 80 MV/m with a gap size of 60 μm . The μ and σ values are the mean and standard deviation of the Gaussian distribution from which the ψ_i values are selected respectively.

Quantity	Value	Unit
P_{Ref}	1E-7	-
Initial E_S	21E6	V/m
E_L	80E6	V/m
γ	0.25	V/m
μ	1	-
σ	0.1	-

Simulations were then performed using different combinations of r_i and h_2 , while r_0 and h_1 were held constant. The objectives were to identify a combination of values that produce a clearly discernible decrease in the number of arcs per unit area in the outward radial direction and to ensure that the flat central region remains sufficiently large to serve as a reference for normalization. Given the electrode's azimuthal uniformity, the surface was meshed using concentric bands. The electric field associated with each concentric band was calculated by evaluating the electrode separation at the band's radial midpoint, thereby yielding a band-averaged field. An illustrative example using a relatively coarse mesh is shown in Figure 6.

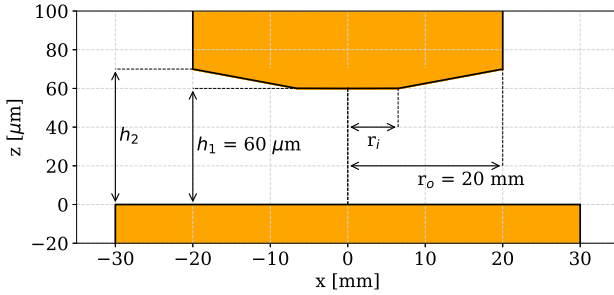


Figure 5: Cross-section of the tapered electrode geometry. Due to the small gap size, units of μm are used on the z -axis, while units of mm are used on the x -axis. Dimensions h_1 and r_0 are shown for completion, but were fixed at 60 μm and 20 mm respectively and not varied in simulation.

In practice, some deviation from this idealized approximation arises due to field enhancement at electrode edges and the presence of fringe fields, particularly at the outermost band. However, arc-induced damage during high-field testing occurs predominantly on the cathode surface, and so the impact of these effects is mitigated by machining the relevant features on the anode rather than the cathode.

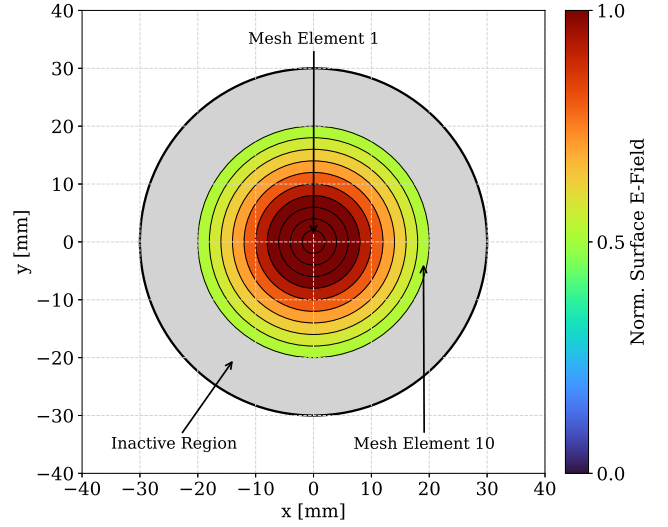


Figure 6: Top-down view of the simulated cathode surface. A total of ten mesh elements are employed in this example, as indicated by the colored (non-gray) bands. The color scale represents the magnitude of the electric field for each element. The field values shown correspond to an arbitrary arrangement in which $h_1 = 60 \mu\text{m}$, $h_2 = 120 \mu\text{m}$, $r_i = 8 \text{ mm}$, and $r_o = 20 \text{ mm}$. As the anode is smaller than the cathode, the surface electric field in the gray region is negligible, and so this area is excluded from the simulation.

The results for three parameter combinations, including the chosen one, are shown in Figure 7. To obtain a more robust result, each parameter combination, i.e., anode geometry, was simulated 100 times, and the results were averaged. The LES control system was included in the model, and a typical conditioning procedure comprised of regular 5 MV/m field increments was employed. The chosen electrode dimensions, following the convention shown in Fig. 5, are summarized in Table II.

Table II: The chosen dimensions for the frustum electrode experiment, following the naming scheme shown in Figure 5. The slope angle is 0.0422° , and results in a field reduction 24 % at the edge of the active area on the cathode.

Quantity	Value	Unit
h_1	60	μm
h_2	70	μm
r_i	6.5	mm
r_o	20	mm

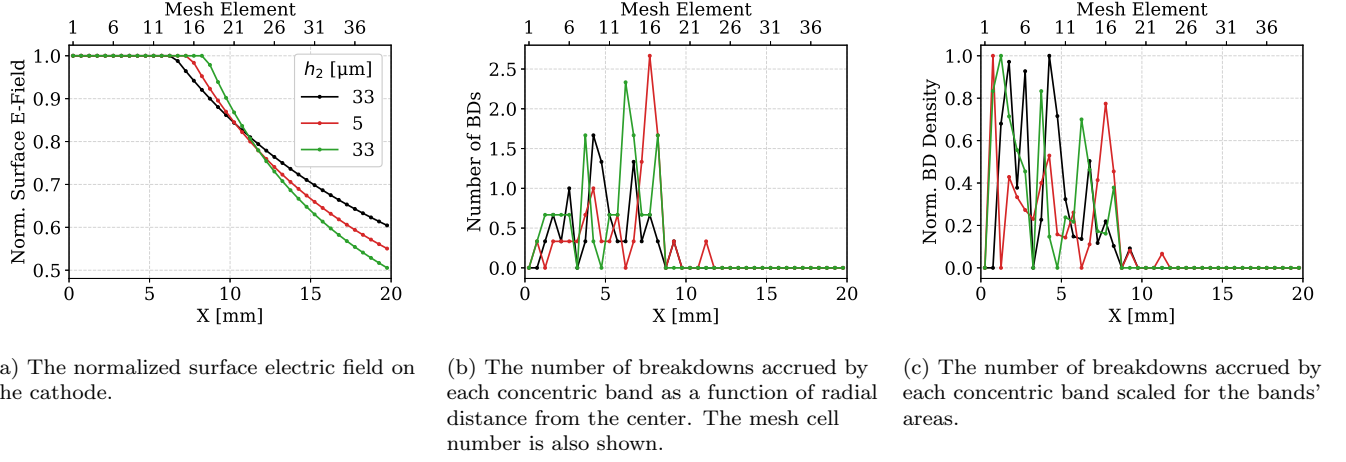


Figure 7: The surface electric field on the cathode (a), the total number of breakdowns accrued by each concentric band (b), i.e., mesh element, and the normalized total number of breakdowns per unit area for each concentric band (c) for simulations of three different anode geometries.

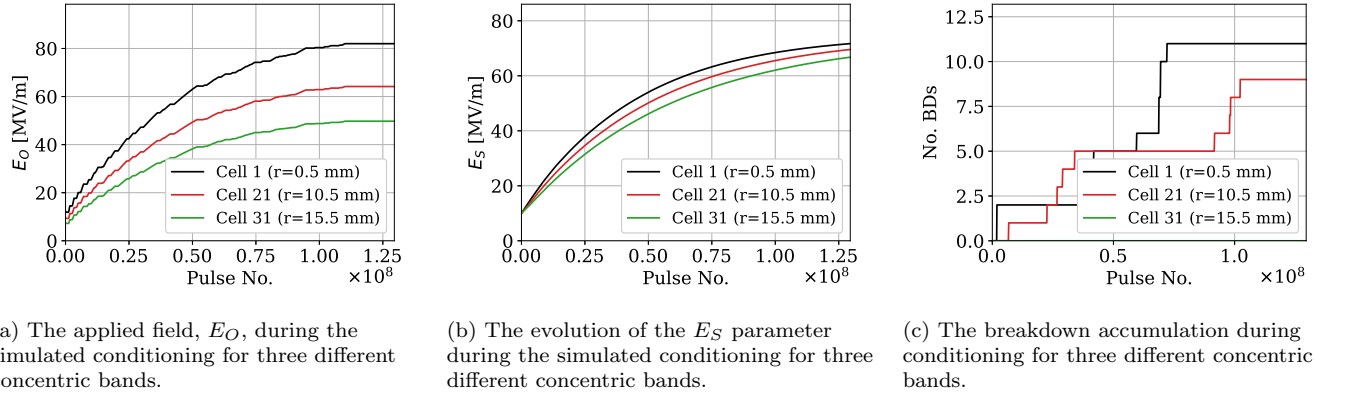


Figure 8: The evolution of E_O (a), E_S (b), and breakdown density (c) for three different concentric bands during the simulation conditioning. Also shown is the point at which they reached XX MV/m, illustrating that different conditioning scenarios produce different....The area of each element was.... **TO BE UPDATED BY LEE. We should show BD density and not E_s maybe...**

C. Predictions of the Model/Emergent Behavior

In Fig. 8, the conditioning of three different mesh cells in the chosen design, each exposed to a different field, is plotted to illustrate the emergent behavior that stems from the assumptions on which the model is founded, namely, that the conditioning effect is induced through applied pulses rather than solely through repeated breakdowns.

The average rate at which the field was increased differed for each cell; hence, each was effectively subjected to a different conditioning procedure. After conditioning, the cells exposed to lower fields exhibit lower final breakdown densities, as expected. The ratios between the applied fields and the final breakdown densities for these elements are X:Y:Z and X:Y:Z, respectively. If the commonly reported E^{30} scaling were applied independently,

a ratio of X:Y:Z between the breakdown densities would be intuitively expected.

In the present model, however, the E^{30} dependence is explicitly incorporated. Despite the difference in applied field, the difference in E_S that emerges is less pronounced, indicating that a significant conditioning effect is consistently produced in all elements. This result can therefore be understood as a consequence of the interplay between the applied field driving the probability of breakdown and producing the conditioning effect, and provides a natural explanation for the breakdown distribution observed in devices that operate with an inhomogeneous surface electric field distribution, such as the cavity shown in Fig. 2.

More notably, as indicated by the vertical lines in Figure 8, the different cells reach a given field level with different breakdown densities. Testing for this behavior

ior is one of the key objectives of the experiment, as it has several implications. Firstly, this result supports the proposal that the conditioning effect is produced by applying high-field pulses, as well as by inducing breakdowns, and provides information on the relative strength of the mechanisms involved. From an operational perspective, it also suggests that devices can be conditioned with fewer breakdowns if less aggressive conditioning procedures are employed, or, for example, if a sacrificial device or a digital twin is operated in parallel to regulate the conditioning. However, at present, it is not known whether such an approach would yield any benefit.

V. EXPERIMENTAL RESULTS - VICTORIA

A. Conditioning and spatial breakdown locations

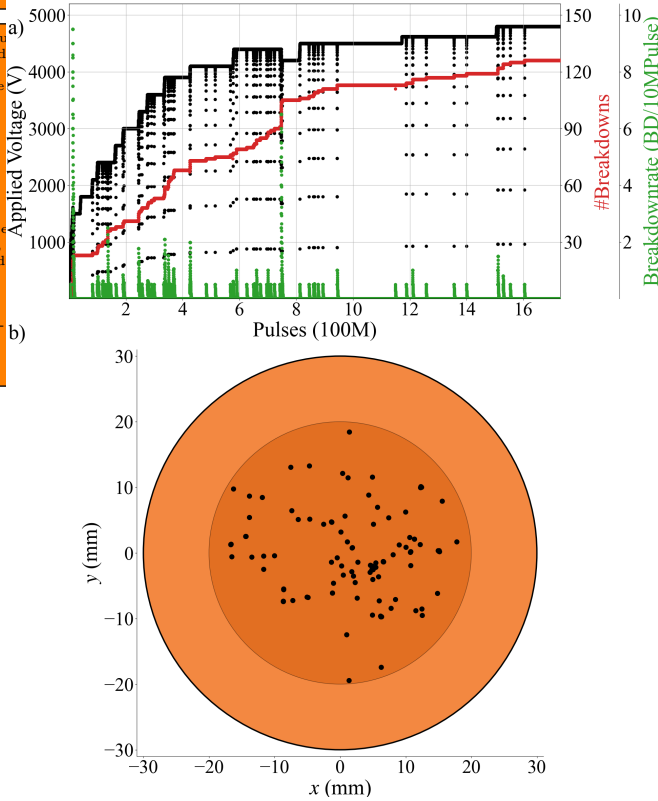


Figure 9: (a) Conditioning curve for the sloped electrode pair. The black line is the applied voltage, the red line is the number of accumulated breakdowns, and the green line is the breakdown rate. The center of the electrode reached $E_{center} = 80$ MV/m after 20 M pulses. (b) Accumulated breakdown locations over the cathode surface. The darker circle on the surface represents the effective field-exposed area, which is equivalent to the anode surface with a radius of 20 mm.

As briefly introduced in Section III, short high voltage pulses of $1 \mu s$ at 1 kHz are applied on the electrodes.

To condition them, the start voltage was set to 300 V (5 MV/m at the center of the electrode) and rose to 1500 V (25 MV/m at the center of the electrode) as the conditioning goal, in 10 V increments per 100 000 pulses. After letting the system run flat to ensure a lower breakdown rate, the conditioning goal was increased by 300 V, and the process was repeated until reaching a final applied voltage of 4800 V, as seen in Fig. 9 a). The black line represents the applied voltage across the electrodes, $V_{applied}$, the red line shows the cumulative number of breakdowns, and the green line corresponds to the BDR.

The surface electric field is calculated from Eq. 2. When the center of the electrode reaches a surface electric field of E_{center} , the outer edge experiences a reduced field strength, given by $E_{edge} = 0.86E_{center}$. Since the center was conditioned to a surface electric field of 80 MV/m, the outermost edge was conditioned to 68.8 MV/m. The spatially located breakdown locations over the cathode surface are shown in Fig. 9 b), where the darker region is the field-exposed area equivalent to the area covered by the anode.

B. Spatial Division of Conditioning

To better understand the relationship between conditioning and breakdown behavior, the electrode surface was divided into three sections: A, B, and C. Spatially resolved breakdown events are categorized according to these sections, as shown in Figs. 10 a-c). Section A (inner) corresponds to the flat region of the anode, which exhibits a uniform surface electric field and covers $r \leq 6.5$ mm. Section B (middle) covers the area from the edge of section A to the midpoint of the slope, given as $6.5 \text{ mm} < r \leq 13.25$ mm. Section C (outer) covers the last part of the exposed electric-field area for $r > 13.25$ mm. Unlike the simulated example, the experimental electrode had fewer breakdowns. Therefore, it was divided into 3 main areas instead of 10.

Figures 10 d-f) present the conditioning data segmented by the spatial regions. In each figure, the blue line represents the spatially averaged surface electric field, the red line shows the cumulative number of breakdowns, and the green line indicates the spatial breakdown density (BDD), which is the number of breakdowns divided by the section's surface area. The maximum (average) surface electric fields achieved in Sections A, B, and C were $E_{inner} = 80$ MV/m, $E_{middle} = 77$ MV/m, and $E_{outer} = 71$ MV/m, respectively.

The number of accumulated breakdowns per section is highest in region B, followed by region C and then region A, due to the increasing surface area as the radius increases. In fact, the corresponding final breakdown densities were 24.0 BD/cm^2 , 12.2 BD/cm^2 , and 5.0 BD/cm^2 for regions A, B and C, respectively.

Figure 11 shows the radial distance as a function of the breakdown number. The earliest breakdowns were predominantly concentrated in section A, followed shortly

The model also grants access to quantities... mention Es or psi parameters here? Or is this too much detail. Probably too much for this paper...

I think you should add a paragraph here to state what the conditioning procedure was i.e. what was algorithmic etc. At the very least, you should explicitly state that a "standard" conditioning procedure was used.

LEE-SHOULD WE HAVE ONE PLOT OF THE ELECTRODE ELSEWHERE IN THE PAPER TO LABEL THE REGIONS? Would it be better to have one plot for each parameter? i.e., having all the applied fields on one plot, all the bds on another... to make comparison easier. We can then draw a line where each region reaches xx MV/m. TO BE DISCUSSED...

Note from Lee - The way the electrode has been divided should be explicitly mentioned. So we should state the middle portion is the 6.5 mm radius and then the step sizes used when dividing it. Then state we normalise for area anyway. Maybe also state that we use three radial bands, since there aren't many breakdowns.

look up this value

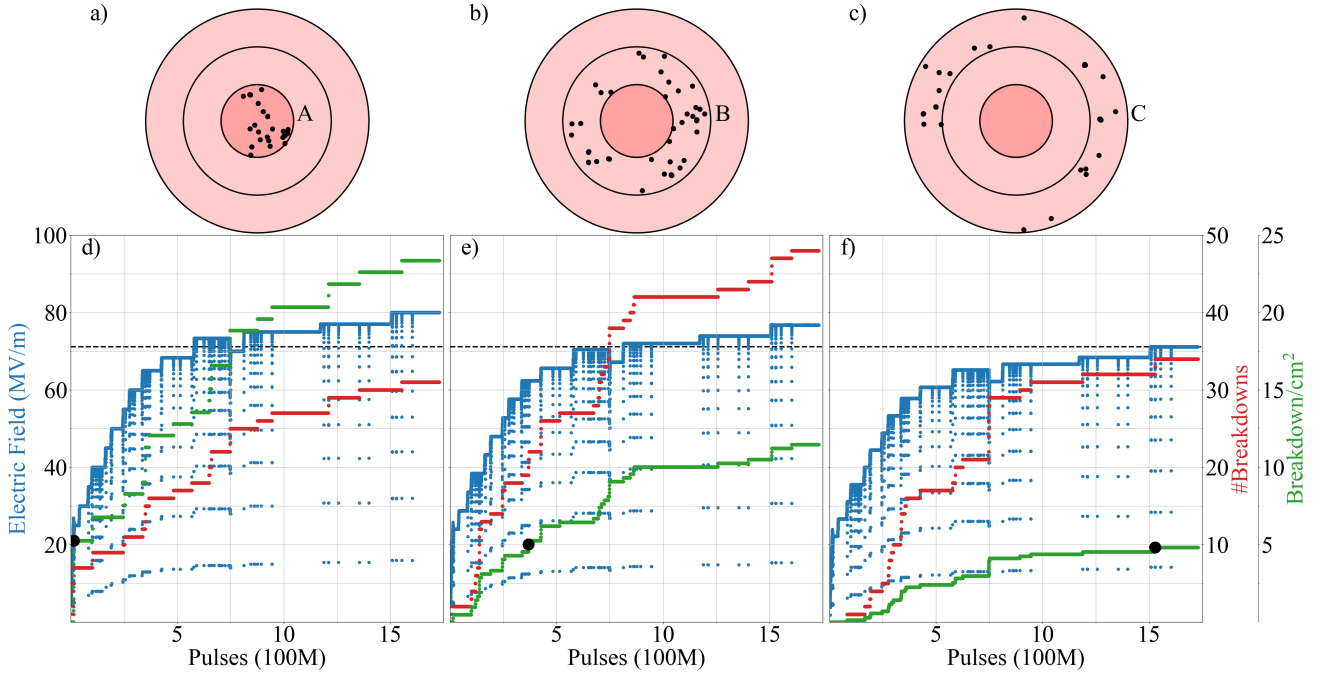


Figure 10: a-c) Spatially located breakdown positions, divided into sections A (inner), B (middle), and C (outer). d-f) Spatially divided conditioning where the blue lines are the average electric field of each region, the red lines are the accumulated breakdowns in each region, and the green lines are the corresponding breakdown density of each region. **LEE- SHOULD WE HAVE ONE PLOT OF THE ELECTRODE ELSEWHERE IN THE PAPER TO LABEL THE REGIONS? Would it be better to have one plot for each parameter? i.e., having all the applied fields on one plot, all the bds on another... to make comparison easier. We can then draw a line where each region reaches xx MV/,m. TO BE DISCUSSED...**

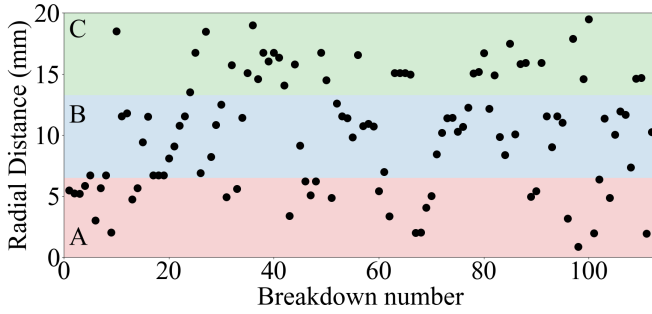


Figure 11: Breakdown number versus the distance from the center of the electrode to the breakdown. Regions A, B and C are shown in red, blue and green, respectively.

by events in section B and then in section C. After approximately 30 breakdowns, the spatial distribution becomes more uniform, with breakdowns occurring across the entire electrode surface, with no spatial correlation observed. This trend is also evident in Figure 10, where the breakdown number in each region increases throughout the conditioning process. However, due to differences in area, the increase in BDD is weaker at greater radial distances.

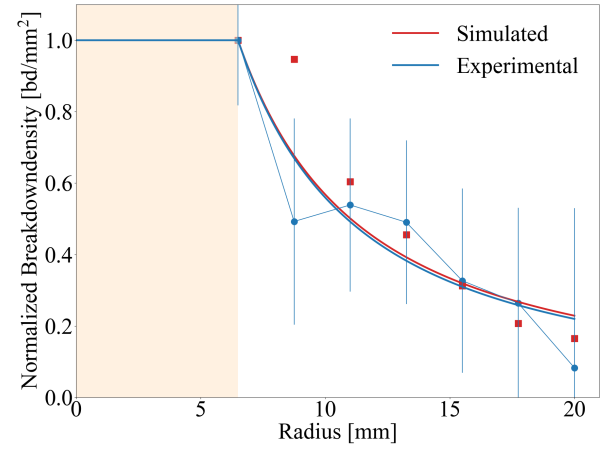


Figure 12: Polynomial fits to BDD over the radius of the electrode. The blue circles and red squares are the discrete experimental and simulated points, respectively. The blue and red lines are the fitted exponential decays. The scaling factor for the experimental data was found to be $\beta_e = 1.35$ and $\beta_s = 1.31$ for the simulated.

C. Simulated results against experimental observations

Because the surface electric field is inversely related to the radial distance as $E \propto 1/r$ when $r > 6.5$ mm and constant for $r \leq 6.5$ mm, the BDD can be found as

$$BDD = \begin{cases} \text{const}, & r \leq 6.5 \text{ mm} \\ \alpha(r - 6.5 \text{ mm})^{-\beta}, & r > 6.5 \text{ mm} \end{cases} \quad (17)$$

where α is a constant, removed by normalization, and β is the scaling factor. Figure 12 shows the experimental and simulated BDDs where the electrode is divided into 7 sections. The best fits of the experimental and simulated decays were found to be $\beta_e = 1.35$ and $\beta_s = 1.31$, respectively.

Since the simulations are configured to record 1000 breakdown events before ending. In contrast, the experiments exhibit a significantly lower number of breakdown occurrences. This results in a lack of statistics for the experimental electrodes and limits the number of sections into which the electrode surface can be divided. In Figure 12, the electrode was divided into 7 sections, where the outer 6 have equal radial distance between them. No change in the results was observed if the area was divided into sections having equal surface area instead.

VI. DISCUSSION

We will now draw the main insights from the experimental data and the corresponding Monte-Carlo simulations. Figure 12 clearly shows the effect of field-dependent conditioning, as the spatial dependence of breakdown density on electric field is X-to-be-recalculated compared to 30 for the dependence of breakdown rate on applied voltage, and from a good agreement between the experimental data and the Monte Carlo simulation.

Here, put in a paragraph that ties back to the Monte Carlo assumptions. Important assumptions are how it is meshed and also that it goes to 1000 breakdowns before stepping (I think).

We can also see that breakdowns start in the highest field region, and progressively work their way out towards the lower field regions as the high field region becomes conditioned and can support higher fields. This, too, is exactly what is expected if a higher field drives the conditioning process more.

These results show that the total number of breakdowns that occur when conditioning to a specific field level depends on how the applied field is ramped. This can be seen in Fig. 10 by considering for the 65 MV/m electric field reached in region C at the end of conditioning. BDDs of , 6.4 and 13.5 BD/cm² occurred reaching 71 MV/m in regions C, B and A respectively. This dependence occurs because the overall breakdown rate of the electrodes is used to control the applied voltage ramp-up

speed in the conditioning algorithm, so the lower field regions are conditioned more slowly. This dependence also supports the previous observations that the number of pulses, rather than the number of breakdowns, condition the device, and extends the observation to local conditioning. [36].

In more general terms, the agreement between Monte Carlo simulation and experiment show that there should be a physical manifestation of E_{state} , either on the electrode surface or in the sub-surface material structures. The results so far do not give a clear insight into what this might be. The dislocation dynamics model, which successfully predicts the dependency of breakdown rate on field and temperature, could fulfill such a mechanism. In this case the dislocation population in the sub-surface would evolve with for example mobile dislocations being pinned as conditioning progresses. Such a process would occur more quickly under the increased tensile stress on the surface due to increased applied field, as required for a physical manifestation of E_{state} .

In previous experiments comparing conditioning between soft and hard copper [57]. Soft copper, also known as heat treated copper, is known to need a longer conditioning time to reach the same field levels as hard copper. On the hard copper, the breakdowns are located on the outer electrode edge, where the surface electric field is at its highest. Since the rest of the surface is experiencing lower field levels, they are not as exposed to breakdowns. The soft copper on the other hand, requires more time to condition. This gives the field enhanced areas an opportunity to condition for longer time and become more resilient to breakdowns, which causes breakdown to occur at other low field regions on the surface, leading to the even distribution observed in the paper.

Both experimental and simulated data show the same general trend in the decay of breakdown density (BDD) as a function of radial distance r . However, the most intriguing aspect is the conditioning process itself. In experiments, the statistical data is limited by the number of breakdowns a single electrode can sustain. In contrast, simulations were run for over 1000 iterations, enabling more robust statistical analysis and a wider spatial distribution of breakdown events. This allows for a finer division of the surface area under investigation. While the experimental and simulated enhancement factors, β_e and β_s , were similar in value, the experimental estimates were subject to large errors and fluctuations. Interestingly, these fluctuations were less pronounced when the surface was divided into 5 to 12 sections instead of 7.

VII. PROPOSED NEW DISCUSSION COMBINED WITH CONCLUSION(?)

We will now draw the main insights from the experimental data and the corresponding Monte Carlo simulations, as we have demonstrated a new approach to testing conditioning and its effects at a local scale. Figure 12

Had some text about the beta being equivalent to 30 etc, which I also saw I should remove.

LEE - IT WOULD BE GOOD IF WE MAKE HERE CONNECTIONS BACK TO MONTE CARLO ASSUMPTIONS AND PARAMETERS.

VICTORIA - WE SHOULD ADD VISUAL GUIDANCE, POINTS OR A LINE, TO FIG. 8 TO MAKE IT EASIER TO SEE THIS POINT.

I'VE COMMENTED OUT THE FOLLOWING TEXT BECAUSE I BELIEVE I'VE INCLUDED THE MAIN POINTS IN THE TEXT ABOVE. BUT PLEASE MAKE SURE THAT I HAVEN'T MISSED ANYTHING.

THE FOLLOWING PARAGRAPH CAN PROBABLY STAY BUT NEEDS WORK SPECIFIC REFERENCES TO FIGURE NUMBERS IS NEEDED. THE READER OF THIS PAPER WILL BE LOST BY THE REFERENCE TO THE EDGE ELECTRIC FIELD BEING HIGHEST FOR EXAMPLE.

I'M NOT SURE ABOUT THIS PARAGRAPH. IT SEEMS TO BE COMPARING THE RELATIVE STATISTICAL ACCURACIES OF THE SIMULATION AND EXPERIMENT, BUT IT'S NOT CLEAR.

clearly shows the effect of field-dependent conditioning by applying electrodes to varying electric fields, we experimentally examined the relationship between the conditioning effect of high-field exposure and breakdowns.

[Monte carlo bit]

As the electrode's center experiences the highest electric field, it conditions more quickly than the surrounding areas. As region A becomes more resistant to vacuum breakdowns, the spatial breakdown distribution broadens, increasing the probability of breakdowns in both regions B and C. The high-field regions become dominant in the conditioning process, influencing it. This becomes evident in Fig. 10, where the average final electric field between regions A and C is less than 10 MV/m, a difference of less than 10 %, while the BDDs for each section differ markedly at the end of the conditioning process. The two innermost regions surpass the final BDD_C at significantly lower fields, with region A surpassing at 20 MV/m and region B at 62 MV/m (indicated by black dots).

In fact, similar observations have been reported for soft and hard copper [57]. Soft copper, also known as heat-treated copper, requires a longer conditioning time than untreated copper. On the hard copper, vacuum breakdowns were concentrated in the high-field regions,

leaving the lower-field regions undamaged. By contrast, on the soft copper electrode, the breakdowns were more uniformly distributed, with more breakdowns occurring in the lower-field regions. This is a direct consequence of the higher field region having more time to condition, becoming more resilient to breakdowns, and is the same result observed in this paper. In fact, current work is underway to investigate subsurface features that can help our understanding of the conditioning process. Clear dislocation activity has been observed on the surfaces of conditioned electrodes and decays spatially as one moves toward lower-field regions.

In more general terms, the agreement between Monte Carlo simulation and experiment shows that there should be a physical manifestation of E_{state} , either on the electrode surface or in the sub-surface material structures. The results so far do not give a clear insight into what this might be. The dislocation dynamics model, which successfully predicts the dependence of the breakdown rate on field and temperature, could serve as such a mechanism. In this case, the dislocation population in the sub-surface would evolve, for example, mobile dislocations being pinned as conditioning progresses. Such a process would occur more quickly under the increased tensile stress on the surface due to the increased applied field, as required for a physical manifestation of E_{state} .

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